

**BEFORE YOU READ — ANSWER FROM MEMORY**

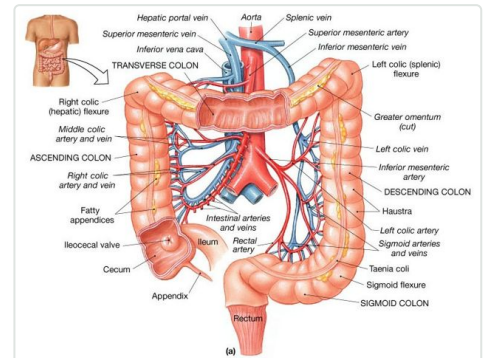
1. Which half of the colon is the absorbing colon, and which half is the storage colon?
2. What triggers mass peristalsis, and how often does it happen?
3. Does the colon secrete digestive enzymes — and what completes the final stage of digestion there?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Regions and the two working halves

- About **1.5 m long, 6.5 cm** in diameter. Regions: **cecum (with appendix)** → ascending → transverse → descending → sigmoid → **rectum** → anal canal.
- The rectum is the **last part of the digestive tract**; it ends at the anal canal, guarded by the **internal and external anal sphincters**.

Half	Job
Proximal “absorbing colon”	Most of the large-intestine absorption
Distal “storage colon”	Stores feces until a “propitious time” for excretion



Regions of the colon, ileocecal valve, haustra.

2 Seven functions

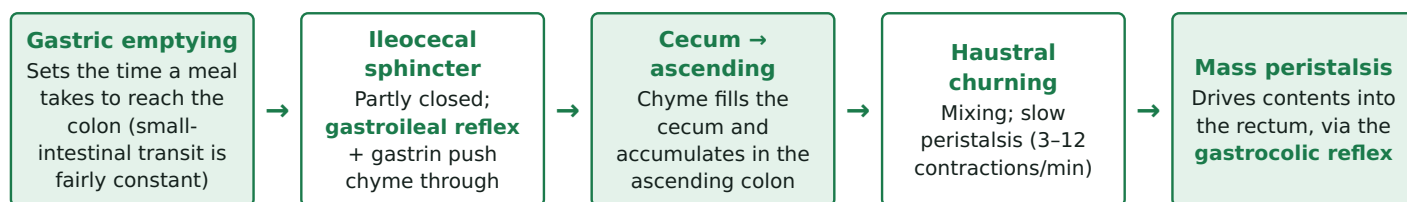
Function	Detail
1 Motility	Haustral churning, peristalsis and mass peristalsis drive the contents into the rectum
2 Bacteria	Convert proteins to amino acids and break these down; make some B vitamins (thiamine B1, B12, folic acid) and vitamin K → contribute to erythropoiesis and blood clotting
3 Absorption	Some water and ions , vitamins, glucose, alcohol, and drugs (anaesthetic agents, sedatives, steroids)
4-5 Feces, defecation	Formation of feces ; emptying of the rectum
6 Excretion	Heavy metals (mercury, lead, bismuth, arsenic) leave in feces
7 Secretion	Mucin , chloride and bicarbonate

3 Mucus secretion

Feature	Detail
Mucosa	Many crypts of Lieberkühn ; no villi ; epithelial cells contain almost no enzymes
Role of mucus	Lubricates the colon and protects the mucosa
Nerve control	Pelvic nerve stimulation ↑ mucus secretion together with ↑ peristaltic motility
Extreme parasympathetic	Bowel movement of ropy mucus as often as every 30 min, with little or no fecal material

4 Motility: from ileum to rectum

VISUAL ANCHOR · HOW A MEAL REACHES THE RECTUM



Movement	What happens
Ileocecal sphincter	Normally partially closed → chyme enters the cecum slowly . After a meal, the gastroileal reflex intensifies ileal peristalsis; gastrin relaxes the sphincter; cecal distension intensifies its contraction
Haustral churning (mixing)	Haustra stay relaxed and distend as they fill; at a certain point the walls contract and squeeze the contents into the next haustrum
Mass peristalsis (propulsion)	Strong wave that begins about the middle of the transverse colon and quickly drives contents into the rectum. Food in the stomach starts it (gastrocolic reflex): 3-4 times a day , during or just after a meal
Transit	Small labelled markers take 36-48 h to pass through the large intestine

5 Chemical digestion: no enzymes — bacteria do the work

Substrate	Bacterial product	Fate
Remaining carbohydrate	Fermentation → hydrogen, carbon dioxide, methane	Flatus (flatulence if excessive)
Remaining protein / amino acids	Indole, skatole , hydrogen sulfide, fatty acids	Some indole + skatole leave in feces (odour); the rest is absorbed → liver → less toxic compounds → urine
Bilirubin	Simpler pigments, including stercobilin	Gives feces their brown colour
Vitamins	Some B vitamins and vitamin K	Absorbed in the colon

6 Absorption and feces formation

Feature	Detail
Feces formation	After 3-10 h in the large intestine chyme is solid or semisolid through water absorption = feces
Feces contain	Water, inorganic salts, sloughed epithelial cells , bacteria and their decomposition products, unabsorbed digested material, indigestible food
Water	90% of all water absorption is in the small intestine, yet the colon matters for water balance: of the 0.5-1.0 L entering, all but ~100-200 mL is absorbed by osmosis
Reabsorbed	Water ; ions (Na^+ , Cl^-); vitamins K , biotin (B7) , B5 ; organic wastes (urobilinogens, stercobilinogens); bile salts ; toxins

P Must-remember pearls

- **Proximal colon = absorbing; distal colon = storage.** The colon has crypts of Lieberkühn, no villi and **no secreted enzymes**; mucus lubricates and protects.
- **Ileocecal sphincter** is partly closed; the **gastroileal reflex** and **gastrin** open the way after a meal; **cecal distension** tightens it.
- **Haustral churning mixes; mass peristalsis propels** — it starts mid-transverse colon, is triggered by the **gastrocolic reflex** and occurs **3-4 times a day** after meals.
- **Bacteria** ferment carbohydrate (H_2 , CO_2 , methane → flatus), make **indole/skatole** (odour), turn bilirubin into **stercobilin** (brown colour) and supply **vitamin K** and some B vitamins.
- Feces form after **3-10 h** of water absorption; all but **~100-200 mL** of the 0.5-1.0 L entering the colon is absorbed by osmosis.

Q Test yourself

1. A student describes haustral churning. Which description is correct?

- A. Contractions of the rectal wall force feces toward the anus
- B. Relaxed haustra distend as they fill, then contract and squeeze the contents into the next haustrum
- C. A strong wave begins mid-transverse colon and drives the contents into the rectum
- D. The haustra contract continuously and empty the colon within minutes
- E. The ileocecal sphincter relaxes and pushes chyme into the cecum

2. A man has a strong urge to open his bowels during a meal. Which reflex explains why mass peristalsis is triggered then?

- A. The gastrocolic reflex, initiated by food in the stomach
- B. The enterogastric reflex, initiated by chyme in the rectum
- C. The vesicointestinal reflex, initiated by bladder irritation
- D. The peritoneointestinal reflex, initiated by irritation of the peritoneum
- E. The gastroileal reflex, initiated by distension of the cecum

3. A patient complains that his stools have a strong odour. Which bacterial products contribute to this?

- A. Stercobilin
- B. Hydrogen and methane
- C. Mucin and bicarbonate
- D. Indole and skatole
- E. Vitamin K and biotin

4. Which pigment produced by bacterial breakdown of bilirubin gives feces their brown colour?

- A. Urobilinogen in the urine
- B. Indole
- C. Hydrogen sulfide
- D. Skatole
- E. Stercobilin

5. After a colon resection, a surgeon wants to preserve the part of the colon that mainly stores feces until an appropriate time for excretion. Which part is this?

- A. The ascending colon only
- B. The transverse colon only
- C. The distal half (the storage colon)
- D. The cecum only
- E. The proximal half (the absorbing colon)

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — Haustral churning: haustra remain relaxed and distend; at a certain point the walls contract and push contents to the next haustrum.

2 — A — Food in the stomach initiates the gastrocolic reflex; mass peristalsis usually occurs 3–4 times a day, during or just after a meal.

3 — D — Some of the indole and skatole produced by bacteria is eliminated in feces and contributes to their odour.

4 — E — Bacteria decompose bilirubin into simpler pigments, including stercobilin, which colours the feces brown.

5 — C — The distal colon functions principally for feces storage (storage colon); the proximal half is the absorbing colon.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam